

Adaptive Self-Consistency for Long-Form Factuality on Long-Tail Facts

Aggregation, saturation, and budget-aware inference

Daniil Moskovskiy Rafael Grigoryan Evgenii Shuranov
Wenshuai Yin Alexander Panchenko

AIRI ■ Skoltech ■ MSU ■ ITMO University ■ HSE

KONVENS 2026

Adaptive Self-Consistency

2026-08-28

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Key message: long-form factuality needs more than a single fluent completion. Transition: first make the failure mode concrete, then show how aggregation and adaptive sampling address it.

Fluent long answers can hide many factual failures

The setting

- Entity-centric questions require **long, coherent answers**.
- Each answer contains many independently checkable claims.
- Rare entities make parametric recall less reliable.

A convincing answer
many atomic claims



Unsupported claims
remain persuasive

long tail

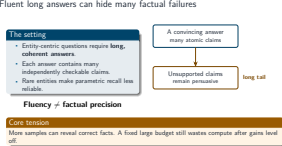
Core tension

More samples can reveal correct facts. A fixed large budget still wastes compute after gains level off.

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Fluent long answers can hide many factual failures

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Key message: the paper treats factuality as a claim-level problem and compute as a variable resource.
Transition: this motivates three empirical questions rather than a new evaluator or benchmark.

Three questions organize the study

1

Does sampling help?
Across head, torso, and tail entities.

2

How should candidates combine?
Synthesis versus voting and selection.

3

Can compute be routed?
Preserve factuality while reducing samples.

Empirical thesis

LLMAgg provides the main factuality gain; adaptive routing compresses a conservative budget, but does not replace calibration.

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└ Three questions organize the study

Key message: keep the talk focused on these three questions. Transition: answer them with a controlled multilingual, multi-domain evaluation.

Three questions organize the study

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Across head, torso, and tail entities.

2
How should candidates combine?
Synthesis versus voting and selection.

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Empirical thesis
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Evaluation: long-form facts across popularity tiers

RiDiC benchmark

- 3 domains: rivers, cars, disasters.
- 2 languages: English and Chinese.
- Wikipedia pageview tiers: **head**, torso, **tail**.
- Public pool: up to 99 completions per input.

Fact-level metric

$$F(y) = \{f_1, \dots, f_n\}$$
$$\text{Fact}(y) = \frac{1}{|F(y)|} \sum_{i=1}^n \mathbf{1}[f_i \text{ supported}]$$

FactOWL extracts atomic facts and verifies them against the supplied Wikipedia context.

Lang.	Domain	Tier	RiDiC entity example
EN	cars	tail	Jiotto Caspita
EN	rivers	head	Blue Nile
EN	disasters	torso	Super Outbreak
ZH	cars	tail	劳斯莱斯 Silver Wraith
ZH	disasters	torso	2024 年花莲地震
ZH	rivers	head	伏尔塔瓦河

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Evaluation: long-form facts across popularity tiers

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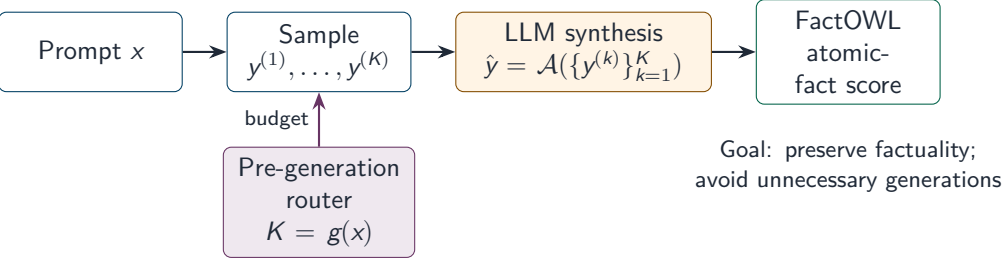
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Same generator: Llama 3.1 8B Instruct

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ZH	rivers	head	伏尔塔瓦河

Key message: the benchmark separates popularity from language and domain, while FactOWL scores precision over atomic claims. The table shows actual RiDiC case entities across those axes. Transition: now show where sampling, synthesis, and routing enter the pipeline.

Method: sample, synthesize, score, then route the budget

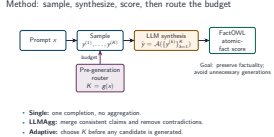


- **Single:** one completion, no aggregation.
- **LLMAgg:** merge consistent claims and remove contradictions.
- **Adaptive:** choose K before any candidate is generated.

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Method: sample, synthesize, score, then route the budget



Key message: the intervention is content-level synthesis plus a separate pre-generation budget policy.
Transition: why synthesis matters becomes clearer when candidate answers disagree.

Why synthesis can beat selecting one answer

	Candidate A	Candidate B	Candidate C
Claim 1	SUV / MG ZS platform	electric SUV	all-electric sedan
Claim 2	1.5L engine	range up to 600 km	unveiled in 2023
Claim 3	manual transmission	Tesla Model Y	600+ horsepower variant

Aggregation objective

Preserve supported, complementary facts; omit unsupported claims and contradictions.

- Selection can only inherit the errors of one existing response.
- LLMAgg writes a new response from the pooled evidence.

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Why synthesis can beat selecting one answer

Key message: this is a precision story, not a recall or oracle-bound story. The best sampled response is a diagnostic; synthesis can produce a stronger new answer. Transition: compare that mechanism with every baseline explicitly.

Why synthesis can beat selecting one answer

	Candidate A	Candidate B	Candidate C
Claim 1	SUV / MG ZS platform	electric SUV	all-electric sedan
Claim 2	1.5L engine	range up to 600 km	unveiled in 2023
Claim 3	manual transmission	Tesla Model Y	600+ horsepower variant

Example: Xiaomi SU7; Candidate B also mentions a 2022 launch.

Aggregation objective
Preserve supported, complementary facts; omit unsupported claims and contradictions.

- Selection can only inherit the errors of one existing response.
- LLMAgg writes a new response from the pooled evidence.

Baselines separate agreement, selection, and synthesis

Condition	What it does	Deployable?
Single	One independently sampled completion	yes
MC	Majority co-occurrence over responses	yes
MF	Majority voting over normalized atomic facts	yes
SP (SeqProb)	Select highest mean token log-probability	yes
Best-of- K logprob	Same selector on the matched $K = 20$ pool	yes
Best sampled	Oracle picks factuality-best candidate after scoring	no
LLMAgg	LLM synthesizes a new response from the pool	yes

Fair comparison

The compute-matched best-of- K selector uses the same 20-sample pool as LLMAgg@20; full-budget results use 99 candidates.

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Adaptive Self-Consistency

Baselines separate agreement, selection, and synthesis

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Fair comparison

The compute-matched best-of- K selector uses the same 20-sample pool as LLMAgg@20; full-budget results use 99 candidates.

Key message: the paper does not compare one vague baseline against one vague method; it isolates response agreement, fact voting, logprob selection, oracle selection, and content synthesis. Transition: the main table shows a consistent ordering.

LLM synthesis is the strongest deployable condition

	Single	Best samp.*	MC	MF	SP	LLMAgg@99
English macro	0.279	0.549	0.128	0.115	0.332	0.578
Chinese macro	0.322	0.634	0.201	0.028	0.394	0.571

*

Across six language-domain cells

- LLMagg is strongest among deployable methods.
- Gains are broad, not driven by one domain.

Interpretation

Synthesis consolidates partially correct material;
next: how quickly does that gain saturate?

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LLM synthesis is the strongest deployable condition

Key message: macro precision rises from 0.279 to 0.578 in English and from 0.322 to 0.571 in Chinese. The gray oracle is not a deployable upper bound because LLMagg writes a new response. Q&A: ZH MF=0.028 is an observed anomaly; the paper does not establish a causal explanation. Avoid presenting a hypothesis about Chinese atomic-fact overlap as a finding. Transition: the large gain arrives early, which creates the budget question.

LLM synthesis is the strongest deployable condition

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* Oracle diagnostic: selected after scaling; not deployable.

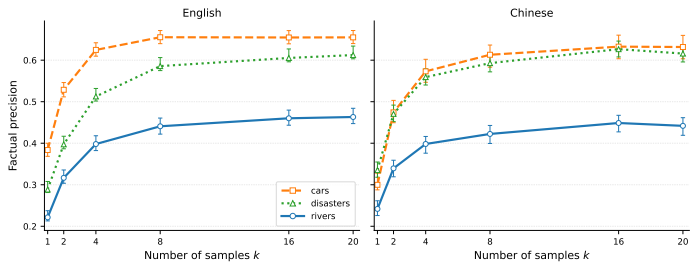
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Synthesis consolidates partially correct material;
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Budget gains level off around $K = 8$ to $K = 16$, with domain exceptions



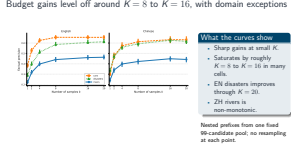
What the curves show

- Sharp gains at small K .
- Saturates by roughly $K = 8$ to $K = 16$ in many cells.
- EN disasters improves through $K = 20$.
- ZH rivers is non-monotonic.

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Budget gains level off around $K = 8$ to $K = 16$, with domain exceptions



Key message: more samples help, but the saturation point depends on domain and language. Transition: if the curves differ, popularity may also remain a structural source of difficulty.

Aggregation improves tail entities, but the head-to-tail gap remains

Lang.	Tier	Single macro Fact	LLMAgg@20 macro Fact
EN	head	0.445	0.828
EN	torso	0.384	0.762
EN	tail	0.243	0.513
ZH	head	0.480	0.757
ZH	torso	0.396	0.675
ZH	tail	0.275	0.501

Stable pattern

Head > torso > tail remains visible after improved inference.

Implication

Aggregation raises precision broadly; long-tail difficulty remains.

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Stable pattern
Head > torso > tail remains visible after improved inference.

Implication
Aggregation raises precision broadly; long-tail difficulty remains.
Values are macro averages over the three domains within each language and tier.

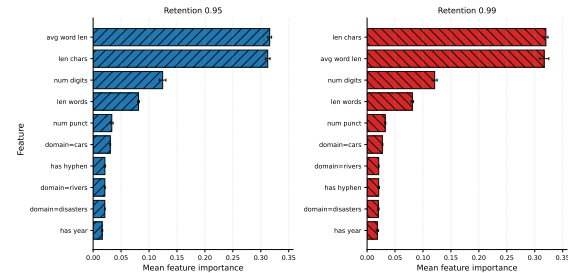
Key message: LLMAgg improves every tier, including tail entities, but does not erase the head–tail gap.
Transition: the remaining question is whether the full 99-sample budget is necessary per input.

Adaptive routing uses pre-generation complexity signals

Controller

Random forest predicts a budget $K = g(x)$ before generation.

- 300 trees; minimum leaf size 3.
- Prompt length, lexical complexity, punctuation, years.
- Language/domain indicators and a domain budget prior.
- No partial output or disagreement features.

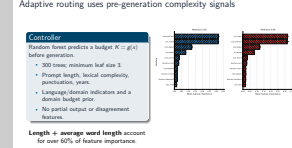


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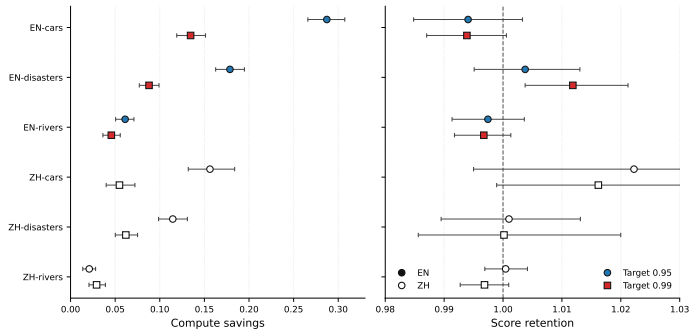
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Adaptive routing uses pre-generation complexity signals

Key message: the router is deliberately simple and pre-generation. It reads input complexity, not model confidence or candidate disagreement. Transition: evaluate the practical tradeoff against the conservative 99-sample reference.



Against $K_{full} = 99$: observed savings, retention stays near 1



Main operating result
2–29%
sampled-generation savings

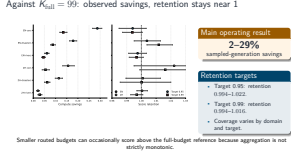
- Retention targets
- Target 0.95: retention 0.994–1.022.
 - Target 0.99: retention 0.994–1.016.
 - Coverage varies by domain and target.

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Against $K_{full} = 99$: observed savings, retention stays near 1

Key message: the controller recovers nearly all full-budget precision while reducing sampled generations, especially in cars and disasters. Transition: this claim is explicitly relative to a conservative reference, not a calibrated optimum.



Adaptive routing saves against $K = 99$, not $K = 20$

$K = 20$ is a different reference

- Budget curves identify $K = 20$ as near-saturated.
- Against $K = 20$, the controller produces no net savings.
- Predicted budgets average 70.6–96.9 samples.

Generalization and safety

- Held-out-domain coverage drops to roughly 0.24–0.65.
- Routing does not detect inputs where LLMagg degrades.
- One generator family; evaluator noise remains possible.

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Adaptive Self-Consistency

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- Predicted budgets average 70.6–96.9 samples.

Savings depend on the reference budget.

Generalization and safety

- Held-out-domain coverage drops to roughly 0.24–0.65.
- Routing does not detect inputs where LLMagg degrades.
- One generator family; evaluator noise remains possible.

Same-distribution routing, not a domain-agnostic recipe.

Key message: the strongest caveat is reference sensitivity. The method compresses $K=99$, but does not beat a hand-calibrated $K=20$ baseline or solve catastrophic aggregation failures. Transition: close by separating what the evidence supports from what remains open.

Three takeaways

- 1 **Synthesis beats selection for long-form factual precision.**
The main gain comes from consolidating complementary facts, not choosing a single “best” completion.
- 2 **Sampling gains saturate early, but saturation is heterogeneous.**
Domain and language determine how much additional computation remains useful.
- 3 **Adaptive routing reduces the $K = 99$ budget; it does not beat $K = 20$.**
It saves 2–29% against 99 samples, while $K = 20$ sensitivity and failure detection remain open.

Answer to the opening tension

Better consolidation matters more than simply generating more answers.

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Adaptive Self-Consistency

Three takeaways

Key message: the paper supports a practical hierarchy: aggregate well first, then calibrate and route the budget. Transition: references, then questions.

Three takeaways

- 1 **Synthesis beats selection for long-form factual precision.**
The main gain comes from consolidating complementary facts, not choosing a single “best” completion.
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Domain and language determine how much additional computation remains useful.
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References

 [D. Moskovskiy et al.](#)
Adaptive Self-Consistency for Long-Form Factuality on Long-Tail Facts. 2026.

 [P. Braslavski et al.](#)
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 [X. Wang et al.](#)
Self-Consistency Improves Chain of Thought Reasoning in Language Models. ICLR, 2023.

 [J. Kim et al.](#)
Reliability-Aware Adaptive Self-Consistency for Efficient Sampling in LLM Reasoning. ACL Findings, 2026.

 [A. Sakhovskiy et al.](#)
FactOWL: A Cost-Efficient Tool for Long-Form Factuality Evaluation. SIGIR, 2026.

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FactOWL: A Cost-Efficient Tool for Long-Form Factuality Evaluation. SIGIR, 2026.

Key message: the primary paper builds on self-consistency and fact-level factuality evaluation, but studies their interaction under long-tail and budget constraints. Transition: thank the audience and invite questions.

Thank you

Questions?

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└ Thank you

Thank you

Questions?

Daniil Moskovskiy | A.Panchenko@skol.tech

Close: return to the practical message: aggregate complementary evidence, then spend compute where it is needed. Invite questions about evaluator reliability, budget calibration, and failure detection.

Backup slides

Additional tables, definitions, and failure cases

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Backup slides

Backup slides

Backup slides

Additional tables, definitions, and failure cases

Use these slides for questions about exact per-cell values, policy definitions, dataset filtering, or qualitative behavior.

Main comparison by language and domain

Lang.	Domain	Single	LLMAgg@99	Gain
EN	rivers	0.211	0.462	+0.251
EN	cars	0.344	0.653	+0.309
EN	disasters	0.283	0.620	+0.337
ZH	rivers	0.266	0.451	+0.185
ZH	cars	0.344	0.632	+0.288
ZH	disasters	0.355	0.631	+0.276

Reading the table

LLMAgg is the strongest deployable method in every language–domain cell; the oracle best-sampled diagnostic is intentionally excluded from this table.

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Reading the table
LLMAgg is the strongest deployable method in every language–domain cell; the oracle best-sampled diagnostic is intentionally excluded from this table.

Use this slide to answer questions about individual cells behind the macro averages.

Supplementary baselines at matched $K = 20$

Lang.	Domain	Single	Best-of- K logprob	ReASC	LLMAgg@20
EN	cars	0.344	0.366	0.354	0.655
EN	disasters	0.287	0.324	0.305	0.612
EN	rivers	0.211	0.221	0.210	0.463
ZH	cars	0.349	0.407	0.349	0.633
ZH	disasters	0.354	0.417	0.368	0.616
ZH	rivers	0.269	0.302	0.277	0.442

Audit note

ReASC is the corrected post-hoc baseline: completion-level FactOWL scores replace the earlier topic-level join.

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Audit note
ReASC is the corrected post-hoc baseline: completion-level FactOWL scores replace the earlier topic-level join.
Best-of- K logprob and ReASC are compared with LLMAgg at the same $K = 20$ budget.

Use this slide when asked whether synthesis still beats stronger selection signals at a matched budget.
ReASC is reported after correcting the topic-level join bug described in the paper.

Adaptive policy: definitions and training protocol

Reported quantities

$$\begin{aligned} \text{compute savings} &= 1 - \frac{\sum_i K_i^{\text{adaptive}}}{N K_{\text{full}}} \\ \text{score retention} &= \frac{\text{Fact}(\hat{y}_{\text{adaptive}})}{\text{Fact}(\hat{y}_{\text{full}})} \\ \text{coverage} &= \Pr[\text{retention meets target}] \end{aligned}$$

Protocol

- Five stratified shuffle splits.
- Test fraction: 30%.
- Split seed: 13.
- Model seed: 13 + fold index.
- Targets: 0.95 and 0.99.

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Adaptive Self-Consistency

Adaptive policy: definitions and training protocol

Reported quantities	Protocol
$\text{compute savings} = 1 - \frac{\sum_i K_i^{\text{adaptive}}}{N K_{\text{full}}}$ $\text{score retention} = \frac{\text{Fact}(\hat{y}_{\text{adaptive}})}{\text{Fact}(\hat{y}_{\text{full}})}$ $\text{coverage} = \Pr[\text{retention meets target}]$	• Five stratified shuffle splits. • Test fraction: 30%. • Split seed: 13. • Model seed: 13 + fold index. • Targets: 0.95 and 0.99.

The reported savings count sampled generations; aggregation prompt growth is not included in that relative measure.

Use this slide to clarify that savings are defined relative to the number of sampled completions, not total aggregation-token cost.

Dataset filtering and FactOWL details

Usable-context counts

	rivers	cars	disasters
English	1000	997	1000
Chinese	723	400	723

FactOWL score

$$F(y) = \{f_1, \dots, f_n\}$$
$$\text{Fact}(y) = \frac{1}{n} \sum_i 1[f_i \text{ supported}]$$

Verification uses the entity's supplied Wikipedia page; no alternative evidence corpus is queried.

Adaptive Self-Consistency

Dataset filtering and FactOWL details

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Filtering requires usable entity titles and supplied Wikipedia contexts.

FactOWL score

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$$\text{Fact}(y) = \frac{1}{n} \sum_i 1[f_i \text{ supported}]$$

Verification uses the entity's supplied Wikipedia page; no alternative evidence corpus is queried.

Use this slide when asked whether the language comparison uses equal sample counts or how the factuality score is constructed.

Qualitative behavior: complementary evidence versus correlated errors

Representative wins

- Jiotto Caspita: +0.415.
- Blue Nile: +0.424.
- Ma Yimoka River: +0.522.

Correct facts are fragmented across candidates; synthesis recovers the shared core.

Representative failures

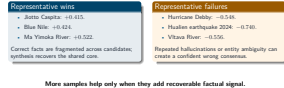
- Hurricane Debby: -0.548.
- Hualien earthquake 2024: -0.740.
- Vltava River: -0.556.

Repeated hallucinations or entity ambiguity can create a confident wrong consensus.

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Adaptive Self-Consistency

Qualitative behavior: complementary evidence versus correlated errors



Use this slide to explain why aggregation is powerful but not uniformly safe, and why the current router does not solve catastrophic failures.